

Ird Data Pipeline Presentation

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Objective

1. To Scrape IRD table data from IRD websites(<https://www.ird.gov.hk/eng/ppr/arc.htm>) and table data from 1-63 (<https://www.ird.gov.hk/eng/ppr/dip.htm>)
2. Store those pdf files to a vector search database so that users can search tax related documents with natural language

Advance Ruling Index

Carrying on a trade, profession or business in Hong Kong

Case No.	Provision of the I.R.O. in respect of which ruling applies	Index
13	section 14	- Whether the non-resident person carried on business in Hong Kong through an agent - Source of profits – trading
16	section 14	- Whether the non-resident person carried on business in Hong Kong through an agent - Source of profits – service income
26	section 14	- Whether carrying on a trade, profession or business in Hong Kong - Source of profits – service income
44	sections 14 and 20(2)	- Whether carrying on a trade, profession or business in Hong Kong - Tax liability of certain non-resident person

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► [Change of accounting date](#)

► [Charter hire income](#)

► [Contracts deemed as leases](#)

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Departmental Interpretation and Practice Notes

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To help keep tax representatives and other interested parties informed of current developments, IRD will advise such persons by e-mail whenever a new DIPN is issued.

For inclusion in the e-mail service mentioned above, registration is required. Please e-mail the registration to taxpr@ird.gov.hk providing the following information -

- name of firm / organisation / individual;
- business registration number, if applicable;
- e-mail address to which IRD e-mail should be sent;
- name and telephone number of contact person.

The following Interpretation and Practice Notes are saved in PDF format. PDF file can be opened by Adobe Acrobat Reader which is available at the [Adobe Systems Incorporated](#).

- Departmental Interpretation and Practice Notes
- Extra Duty Interpretation and Practice Notes
- Stamp Office Interpretation and Practice Notes

No.	Departmental Interpretation and Practice Notes	Date Issued
1.	Profits tax Part A: Computing assessable profits Part B: Revenue recognition Part C: Measurement of inventories or stock	September 2002
2.	Profits tax Part A: Industrial building allowances Part B: Commercial buildings allowances	April 1999
3.	Profits tax - Apportionment of expenses	July 2008
4.	Lease premiums / Non-returnable deposits / Key or tea money / Construction fees etc.	February 2006
5.	Profits tax deductions for expenditure on - Technical education Building refurbishment Prescribed fixed assets Environmental protection facilities	April 2019
6.	Interest Revenue Ordinance - Provisions as to (A) Objections to the Commissioner (B) Appeals to the Board of Review (C) Appeals to the Courts	November 2016

Solution

1. I built a RAG pipeline with Prefect pipeline orchestration python package to streamline the scraping pdf documents, cleansing the markdown files after pdf conversion and vector search using opensearch
2. Prefect Cloud provides a convenient UI for controlling and monitoring the pipelines. Opensearch UI could let users to tests the searching performance



Web scraping



Data Cleansing



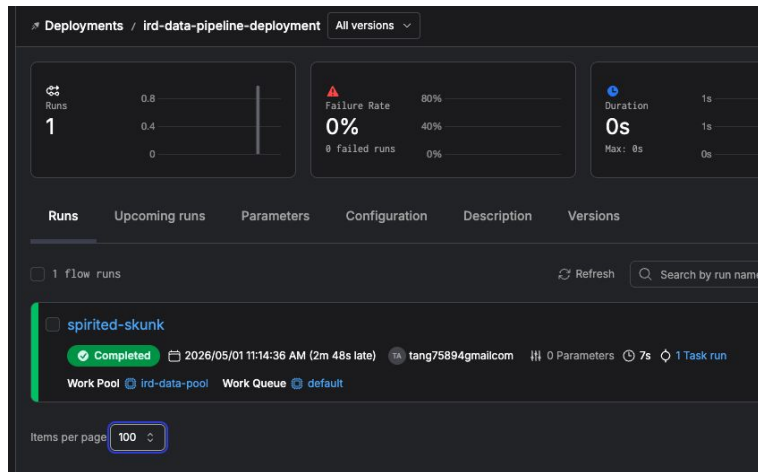
RAG



Web Scraping Flow

1. Scrapy and BeautifulSoup python packages were used to make sure all documents are downloaded
2. Monitor the scraping process in Prefect Cloud UI with logs

The screenshot displays the Prefect Cloud interface for a specific flow run. On the left, a sidebar shows the project structure with 'irid_scrape_data_dag' and 'spirted-skunk' selected. The main panel shows the execution logs for the 'spirted-skunk' flow. The logs indicate that the flow was successfully executed, with a status of 'Completed' and a duration of 4m 21s. The logs also show the flow's entry point, the file path for the 'irid_scrape_data_dag' script, and the successful completion of the flow run.



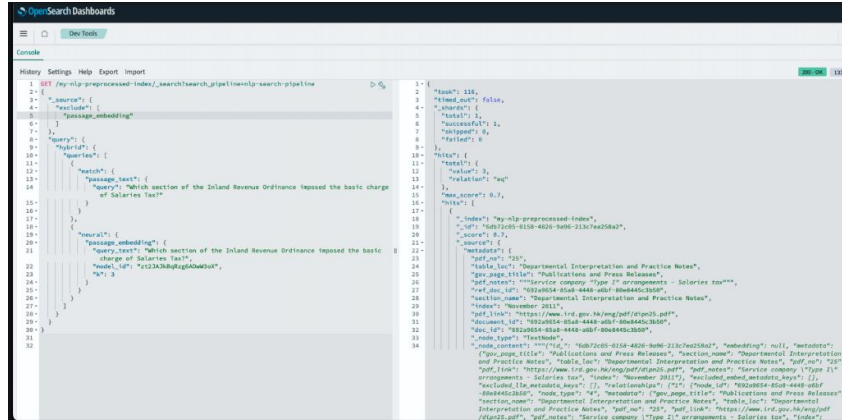


Data Cleansing Flow

1. It involves file conversion from pdf to markdown where llm models are good to process markdown data
2. Human-in-the-loop is needed to check the quality of the markdown files(eg. Any missing words, incorrect format, etc)

RAG Flow

1. It uses Opensearch for RAG vector search. It can perform hybrid search with native features to simplify the RAG pipeline, reducing the need for external, custom code
2. Opensearch has explain API to tune search relevance by showing why a document was ranked highly
3. There is a UI where developers can test the search result with natural language



The screenshot displays the OpenSearch Dashboards interface, specifically the Console tab. The left pane shows a search pipeline named 'my-mlp-preprocessed-index-search-pipeline' with a query that uses a hybrid search strategy. The right pane shows the results of the search, including a list of documents and a detailed explanation of the search results.

```
1 GET /my-mlp-preprocessed-index/_search?search_pipeline=my-mlp-preprocessed-index
2 {
3   "source": {
4     "query": {
5       "hybrid": {
6         "queries": [
7           {
8             "type": "keyword",
9             "field": "text",
10            "query": "Which section of the Inland Revenue Ordinance imposed the basic charge of Salaries Tax?"
11          },
12          {
13            "type": "vector",
14            "field": "text_embedding",
15            "query": "Which section of the Inland Revenue Ordinance imposed the basic charge of Salaries Tax?"
16          }
17        ],
18        "weights": [
19          0.5,
20          0.5
21        ]
22      }
23    },
24    "highlight": {
25      "fields": [
26        "text"
27      ],
28      "fragment_size": 200
29    }
30  }
31}
```

The results pane shows a list of documents, with the first document being a 'Departmental Interpretation and Practice Note' (DIPN) titled 'Salaries tax - arrangements - Salaries tax'.

```
1 {
2   "index": "my-mlp-preprocessed-index",
3   "type": "my-mlp-preprocessed-index",
4   "score": 0.7,
5   "source": {
6     "query": {
7       "hybrid": {
8         "queries": [
9           {
10            "type": "keyword",
11            "field": "text",
12            "query": "Which section of the Inland Revenue Ordinance imposed the basic charge of Salaries Tax?"
13          },
14          {
15            "type": "vector",
16            "field": "text_embedding",
17            "query": "Which section of the Inland Revenue Ordinance imposed the basic charge of Salaries Tax?"
18          }
19        ],
20        "weights": [
21          0.5,
22          0.5
23        ]
24      }
25    },
26    "highlight": {
27      "fields": [
28        "text"
29      ],
30      "fragment_size": 200
31    }
32  }
33}
```

The detailed explanation shows the search results for the first document, including the document ID, score, and the highlighted text.

```
1 {
2   "index": "my-mlp-preprocessed-index",
3   "type": "my-mlp-preprocessed-index",
4   "score": 0.7,
5   "source": {
6     "query": {
7       "hybrid": {
8         "queries": [
9           {
10            "type": "keyword",
11            "field": "text",
12            "query": "Which section of the Inland Revenue Ordinance imposed the basic charge of Salaries Tax?"
13          },
14          {
15            "type": "vector",
16            "field": "text_embedding",
17            "query": "Which section of the Inland Revenue Ordinance imposed the basic charge of Salaries Tax?"
18          }
19        ],
20        "weights": [
21          0.5,
22          0.5
23        ]
24      }
25    },
26    "highlight": {
27      "fields": [
28        "text"
29      ],
30      "fragment_size": 200
31    }
32  }
33}
```



Terraform deployment for AWSOpenSearch and S3

1. Configure [main.tf](#) for creating AWSOpenSearch and S3 services
2. Create services by running commands ``terraform init``, ``terraform plan -out=tfplan`` and ``terraform apply tfplan``
3. Run ``terraform destroy`` if decommission those resources



Prefect Cloud

1. It manages all the flows in a single portal. We can monitoring the deployment process and healthiness of a particular deployment
2. It also manages secrets so no need to do extra works integrating secret values to Prefect environment which is safer.

☐	Name	Last run	Next run	Deployments
☐	ird_preprocess_data_dag Created: 2025/10/04 02:00:13 PM	smiling-cuckoo		1 Deployment
☐	ird_rag_dag Created: 2025/10/04 09:18:51 PM	premium-robin		None
☐	ird_scrape_data_dag Created: 2025/09/02 11:19:20 PM	arcane-crow		1 Deployment
☐	my-flow Created: 2025/09/02 11:16:00 PM	hilarious-tiger		None

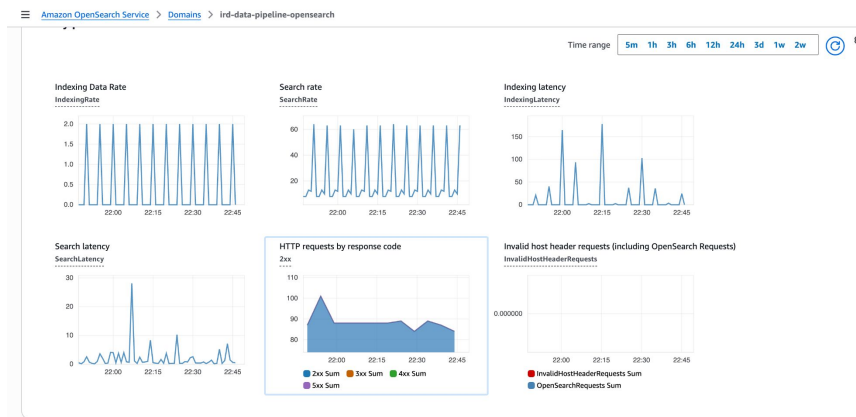
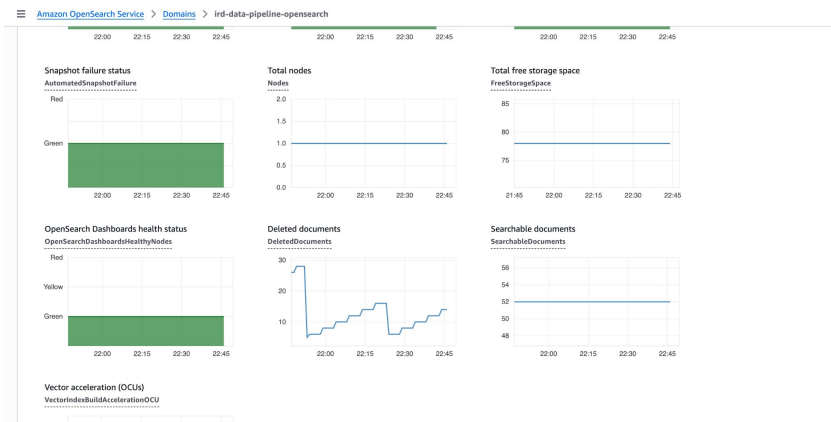
Items per page: 10

☐	Block
☐	aws-access-key-id Secret
☐	aws-opensearch-endpoint Secret
☐	aws-opensearch-password Secret
☐	aws-opensearch-username Secret
☐	aws-secret-access-key Secret
☐	docs-ird-case JSON



AWS Opensearch Service

1. The codes support selection on different vector search tools(chromadb, local Opensearch and AWS Opensearch Service)
2. With AWS Opensearch Service and Prefect Cloud, all the code infrastructure will be setup in cloud





Phase 2 enhancement

1. Setup a chat model with ollama + interface for user to interact with the RAG conveniently
2. Setup terraform to deploy hardware
3. Setup infisical for secrets management since it provides higher security
4. Setup tailscale for vpn to make sure only my laptop can access the application
5. Setup time series database such as Prometheus and Grafana dashboard for system health monitoring
6. Create a opensearch container and deploy to Digital Ocean for long term demonstration purpose